

DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

NAOH-653

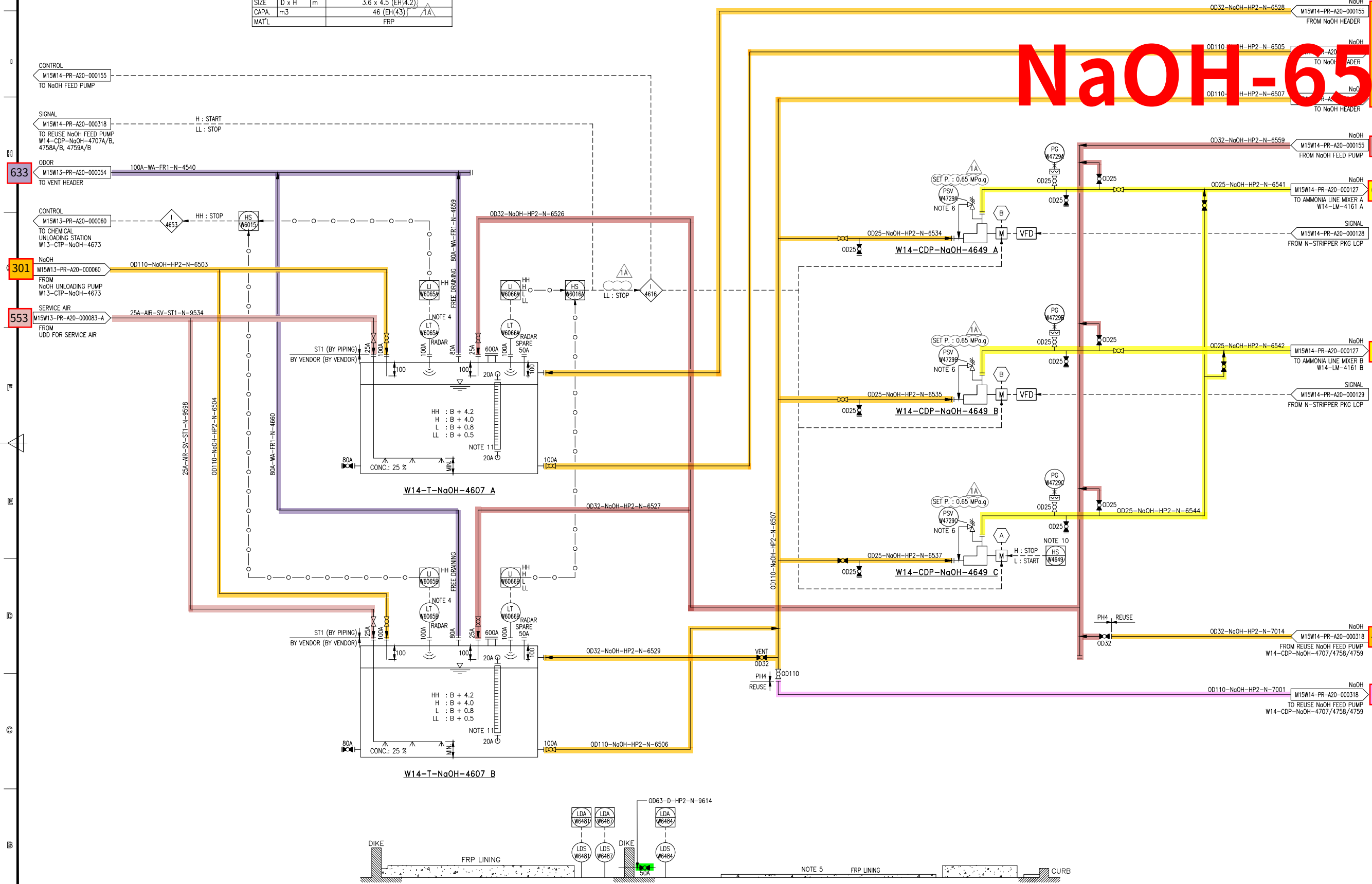
No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	653	4FA	OD32-NaOH-HP2-N-6526-01	1 OF 1	0	CJ-M15W14-PR-A20-000154		
2	653	4FA	OD32-NaOH-HP2-N-6527-01	1 OF 1	0	CJ-M15W14-PR-A20-000154		
3	653	4FA	OD32-NaOH-HP2-N-6559-01	1 OF 2	0	CJ-M15W14-PR-A20-000154		
4	653	4FA	OD32-NaOH-HP2-N-6559-02	2 OF 2		CJ-M15W14-PR-A20-000154, 155		
5								
6								
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9								
10								
11								
12								

ITEM NO.	W14-T-NaOH-4607 A/B
SERVICE	NaOH STORAGE TANK
Q'TY	2
TYPE	VERTICAL CYLINDRICAL
SIZE	ID x H m 3.6 x 4.5 (EH(4.2))
CAPA.	m3 46 (EH(43))
MAT'L	FRP

대외비

KEY PLAN (4F)

NaOH-653



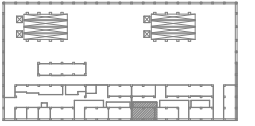
1. PUMP VENDOR 공급.
2. DIAPHRAGM PUMP MAT'L 표기 : HEAD / DIAPHRAGM / CHECK BALL
3. 역류종류에 따라 LINE 색상을 구분하여야 한다.
4. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
5. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAINAGE 인단의 TRENCH로 흡수되어, DRAIN HEADER로 이용된다.
6. INTERLOCK NO. I-4653 (W13-CTP-NaOH-4673 AUTO STOP) : REMOTE STATUS에서 W14-T-NaOH-4607 A/B LEVEL (U-W6065 A/B) HIGH HIGH STOP.
7. INTERLOCK NO. I-4616 (W14-CDP-NaOH-4645 A/B/C, W14-CDP-NaOH-4646 A/B/C, W14-CDP-NaOH-4647 A/B/C, W14-CDP-NaOH-4648 A/B/C, W14-CDP-NaOH-4649 A/B/C AUTO STOP : REMOTE STATUS에서 W14-T-NaOH-4607 A/B LEVEL (LI-W6066 A/B) LOW LOW STOP. ALL SPARE PUMP는 pH SET VALUE HIGH STOP.
8. N-STRIPPER PACKAGE TRAIN A/B 중 선택연전.
9. TANK의 역만 레벨 경계치는 VENDOR에서 공급한다.
10. 모든 가성소다 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
11. 가성소다 환관 AREA에는 비산방지막을 설치한다.
12. OPERATING CAPACITY : 0.5 LPM OPERATING PRESSURE : 0.59 MPa

ITEM NO.	W14-CDP-NaOH-4649 A/B/C
SERVICE	AMMONIA STRIPPING TOWER NaOH FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	11/min.ea x MPa (MAX. 0.52 x MAX 1.0 NOTE 14)
MAT'L	PVDF/PTFE/CERAMIC
POWER kW	(0.4(VWF) / 0.55(BLDC))

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정 EV.	실명 DESCRIPTION	날짜 DATE
DRAWN BY	이인학	
CHECKED BY	한진희	
APPROVED BY	박동영	
제작사	SK 에코플랜트	
공사명	M15X Ph4 WWT Project	
도면명	PIPING & INSTRUMENT DIAGRAM NaOH STORAGE TANK & FEED PUMP (1/2)	
축척	NONE	
날짜	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호	CJ-M15W14-PR-A20-000154	2A

NaOH-653

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KEY PLAN (4F)

NOTE

1. 운전 시 CLOSE 되는 배관에는 LOTO 실시.
2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL
3. 역류종류에 따라 LINE 역상을 구분하여야 한다.
4. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이송된다.
5. PUMP VENDOR 공급.
6. NaOH LINE에 위치한 VALVE에 LOTO 적용.
7. HS-W1001A, W1001B 중 선택은전.
8. AIC-W2254, W2264 중 선택은전.
9. HS-W1902, W1903 중 선택은전.
10. HS-W1202, W1203 중 선택은전.
11. 모든 가정소터 배관의 LEAK POINT에는 비산 방지를 SHIELD 및 SENSING TAPE 설치할 것.
12. 가정소터 펌프 AREA에는 비산방지력을 설치한다.
13. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.59 MPa
14. OPERATING CAPACITY : 8 LPM
OPERATING PRESSURE : 0.59 MPa
15. OPERATING CAPACITY : 7 LPM
OPERATING PRESSURE : 0.59 MPa

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	설명 DESCRIPTION	날짜 DATE

제 DRAWN BY	이 인 학
검 CHECKED BY	한 진 희
인 APPROVED BY	박 동 영
제 작 사	

SK 에코플랜트

공 사 명
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
NaOH STORAGE TANK & FEED PUMP
(2/2)

속
SCALE
NONE

날
DATE
2025. 08. 14

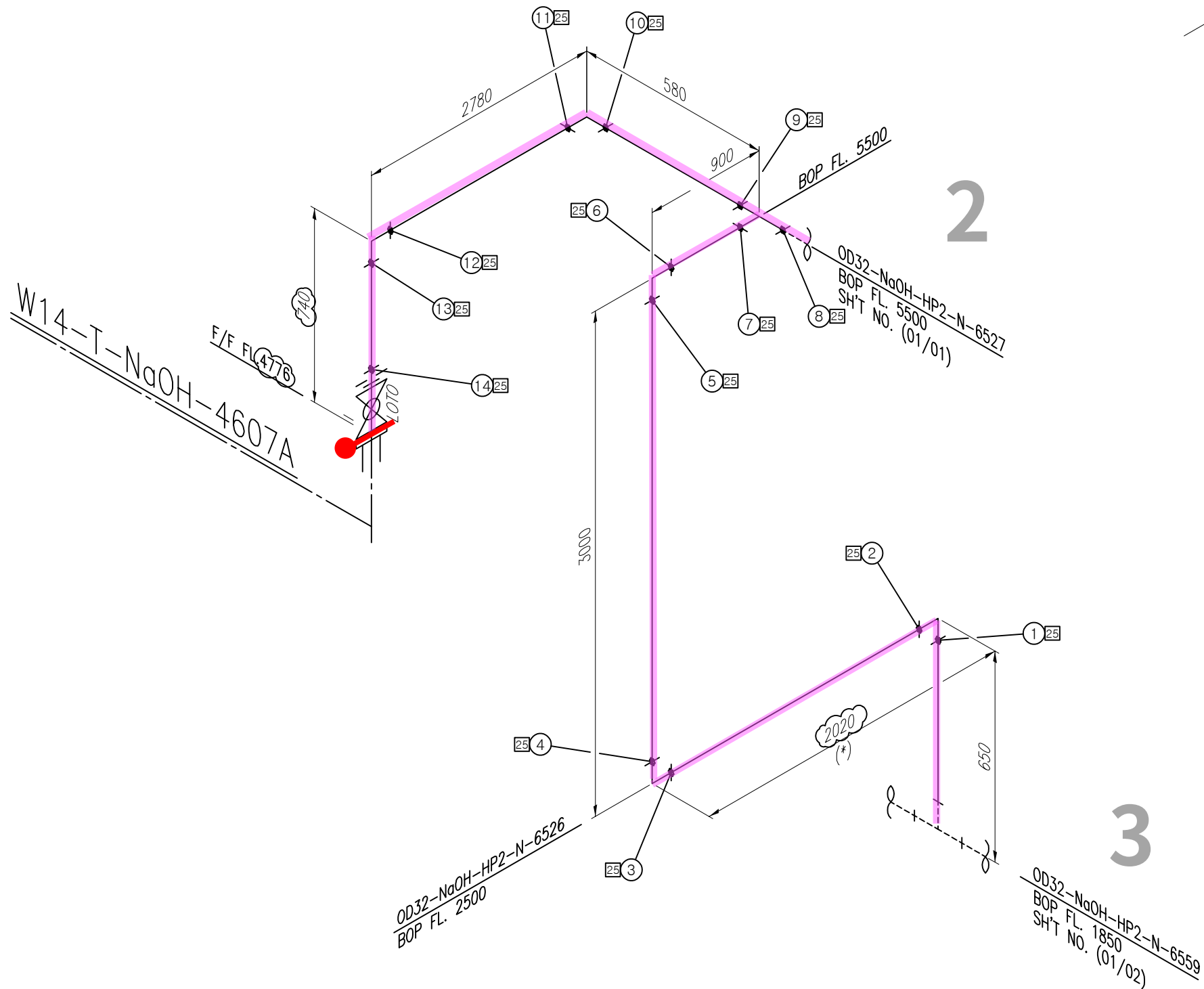
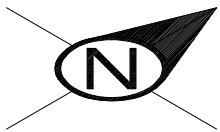
PROJECT NO. 22330D
도면번호
SHEET NO. CJ-M15W14-PR-A20-000155 R1 P&ID FOR NaOH

ITEM NO.	W14-CDP-NaOH-4645 A/B/C
SERVICE	W.W NaOH FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 13)
MAT'L	PVDF/PTFE/CERAMIC
POWER kW	(0.75)

ITEM NO.	W14-CDP-NaOH-4646 A/B/C
SERVICE	CIF 2'nd NaOH FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 14)
MAT'L	PVDF/PTFE/CERAMIC
POWER kW	(0.75)

ITEM NO.	W14-CDP-NaOH-4647 A/B/C
SERVICE	HF 2'nd NaOH FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 15)
MAT'L	PVDF/PTFE/CERAMIC
POWER kW	(0.75)

ITEM NO.	W14-CDP-NaOH-4648 A/B/C
SERVICE	CIF ORGANIC NaOH FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 13)
MAT'L	PVDF/PTFE/CERAMIC
POWER kW	(0.75)



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	6.6	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	6.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040012-D			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	NaOH	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000154			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. OD32-NaOH-HP2-N-6526	SH'T NO. 01 / 01	REV. 1
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

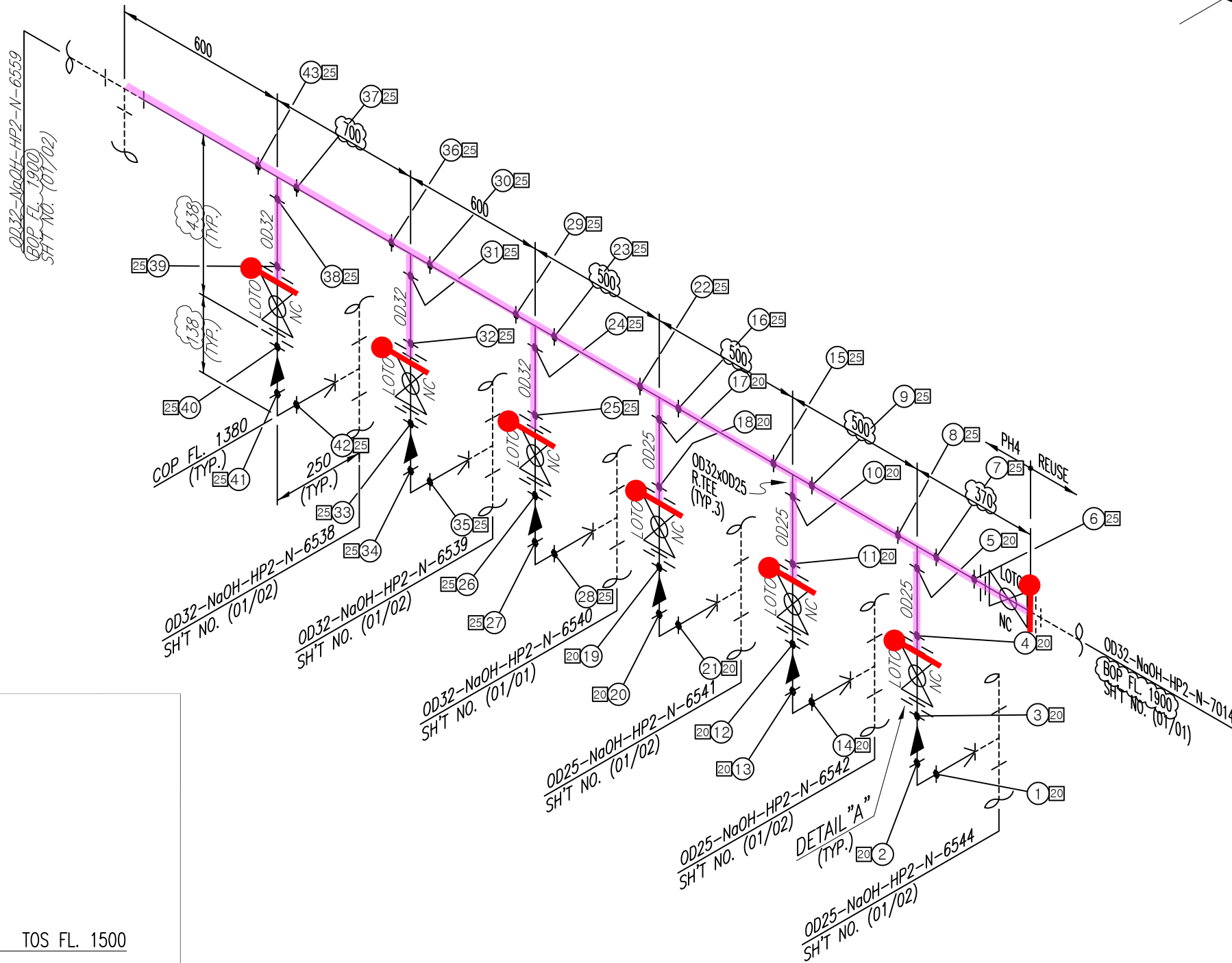


	CONTRACTOR		PROJECT NAME	
			M15X Ph4 Project	
FO	COMMODITY	NaOH	JOB NO.	
FO			22330D	
D	LINE NO.	0D32-NaOH-HP2-N-6527	SH'T NO.	REV.
			01 / 01	1

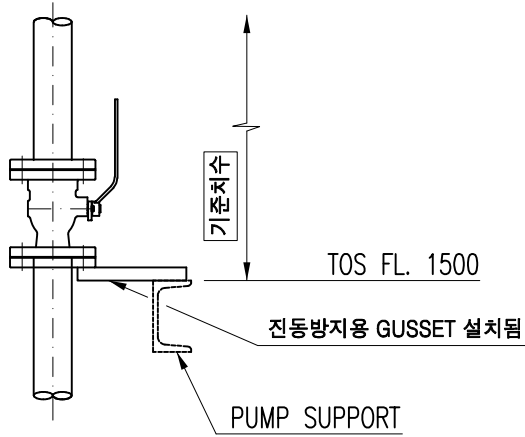
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA W14						
	THICK.	-	DESIGN	6.6	60	UNIT 4FA						
			OPERATING	6.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040012-D		25.03.31	REVISED AS MARKED	황래중	안남후	박동
						REF. P & ID CJ-M15W14-PR-A20-000154		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

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3



DETAIL "A"



도면상의 치수는 상기 GUSSET 하부면 기준치수이니 참조할것.

- <NOTE>
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2. (*) FIELD ADJUST

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* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	CONTRACTOR 	PROJECT NAME M15X Ph4 Project
	THICK.	-	DESIGN	6.6	60	UNIT	4FA								
			OPERATING	6.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040012-A								COMMODITY NaOH	JOB NO. 22330D
						REF. P & ID CJ-M15W14-PR-A20-000155								LINE NO. OD32-NaOH-HP2-N-6559	SH'T NO. 02 / 02
															REV. 0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.